

Delay-Doppler Domain Feature Extraction for Explainable Wi-Fi Activity Recognition via SVM

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Abstract—Joint Communication and Sensing (JCAS) activity recognition applications often operate directly on the Channel State Information (CSI) time series through Neural Networks. This work proposes the use of the Symplectic Fourier Transform coupled with a linear Support Vector Machine classifier, yielding a computationally light and explainable methodology. CSI samples are mapped onto the Delay-Doppler (DD) domain, where the variations in the propagation environment due to human motion manifest as changes in multipath components and Doppler shifts with direct physical interpretation. From short DD-domain CSI blocks, we extract 11 features describing energy distribution, delay and Doppler statistics, entropy, and dominant scattering components. The resulting low-dimensional feature vectors are classified by a linear SVM. The methodology is validated on two independent IEEE 802.11ax Wi-Fi datasets collected during normal indoor activities under different Line of Sight conditions. Results show that DD-based feature extraction combined with a linear SVM achieves consistently high activity recognition accuracy in real life scenarios, even when relying solely on CSI extracted from standard Wi-Fi beacon frames. The approach proves robust across frame rates and propagation conditions while maintaining low computational complexity and full physical interpretability. Moreover, it yields better results than state of the art AI-based classification over the same experiments. The results support the feasibility of lightweight, explainable JCAS applications using commercial Wi-Fi hardware without dedicated sensing transmissions, posing DD feature extraction as a competitor for Deep Learning-based CSI processing.

I. INTRODUCTION

Joint Communication and Sensing (JCAS) is becoming a core component of modern wireless networks. With the promise of seamless integration of the two tasks within the same infrastructure and resources, both academia and industry have made research efforts to study and implement JCAS capabilities in latest generations of wireless networks. Standardization bodies are also moving in this direction: IEEE has approved a Task Group to formalize Wi-Fi sensing [1], and 3GPP considers JCAS a staple for beyond 5G communications [2].

The cornerstone of JCAS is the Channel State Information (CSI), namely the receiver estimation of the time-varying channel frequency response. The CSI is extracted at the receiver from known Orthogonal Frequency Division Multiplexing (OFDM) symbols, either embedded in the frames preamble in packet-based channels – like Wi-Fi – or inserted as pilot signals in the time-frequency Resource Elements grid of circuit-based channels – like in cellular networks. In both cases, the CSI provides per-subcarrier complex channel frequency responses that capture multipath and environmental dynamics. For instance, human motion impacts the channel

by introducing propagation delays and Doppler shifts, leaving measurable signatures on the CSI.

Despite the clear association between the physical structure of the environment and the information embedded in the CSI, most existing approaches interpret this information using Neural Networks (NNs), treating CSI's as high-dimensional inputs and relying on data-driven feature learning. NNs are extremely sensitive to variations in the CSI, enabling fine-grained sensing with astonishingly high accuracy. However, these approaches typically do not look into which CSI features are actually responsible for such accurate performance, and are often difficult to scale from labs to real life [3]. Although some works attempt at a stochastic characterization of the CSI's [4], [5], [6], the focus on Deep Learning (DL)-based processing has largely overshadowed the investigation of the physically meaningful information embedded in them, which could instead support the development of more robust and interpretable models.

This paper adopts a two step methodology. Instead of directly processing the CSI time series, we exploit its transformation into the delay-Doppler (DD) domain through the Symplectic Fourier Transform (SFT) and then separate activities with a linear Support Vector Machine (SVM). The physically interpretable DD representation enables reasoning about the effects of environment dynamics and the constraints imposed to sensing tools by bandwidth and sampling rate. We validate the proposed methodology on two independent IEEE 802.11ax datasets consisting of about 2.5 hours of CSI collections conducted in real-life conditions.

The main contributions of this work are:

- The introduction of the SFT in the context of JCAS, with particular attention to the discrete-time sampling inherent in Wi-Fi and to the delay and Doppler resolution bounds deriving from the channel physical properties;
- The identification of a compact set of physically grounded features that capture the informative properties of the DD-domain representation of CSI samples;
- The design and implementation of a linear SVM classifier leveraging these feature vectors to perform activity recognition;
- Comparison of performance results with a state of the art, AI-based classification technique over the same dataset, showing that the proposed interpretable SFT-based SVM classification performs consistently better.

By combining DD domain analysis with feature-based supervised classification, this work contributes to assessing which physically meaningful properties of the wireless channel drive activity recognition accuracy. The results show that explainable models leveraging propagation theory can compete with data-intensive DL methodologies, opening to new approaches to JCAS in commercial Wi-Fi networks.

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II. RELATED WORK

To contextualize this paper in the vast literature on JCAS, we focus on three aspects relevant for this work, namely Doppler, SVM-based approaches to activity recognition, and their contrast with DL methods. Comprehensive surveys on JCAS can be found in recent tutorials like [7], [8].

A. Doppler analysis for Activity Recognition

Human motion alters the length of the reflected propagation signal paths, inducing Doppler frequency shifts that can be extracted from the CSI variations. Systems like IndoTrack [9] and Widar3.0 [10] exploit the Doppler frequency shift to estimate velocities and trajectories, generating Doppler spectrograms that are unique, activity-specific signatures. However, extracting a reliable Doppler signal from commercial off-the-shelf Wi-Fi devices is challenging due to hardware imperfections that induce CSI phase noise.

Recent paradigms, such as the Time-domain Difference of CSI (TD-CSI) [11], attempt to mitigate these issues by using CSI differencing to isolate high-precision motion information from static reflections. Alternatively, some systems use multi-point two-way communication schemes to cancel devices synchronization errors and phase offsets, enabling reliable Doppler sensing even on single-antenna devices [12].

The framework proposed in [13] and extended in [14] introduced an environment-independent Doppler-based activity recognition tool. The Doppler spectrograms are fed to a Convolutional Neural Network (CNN) trained for cross-environment and cross-subject generalization. The emphasis of the work is on CSI preprocessing to extract clean Doppler information and on integrating Multiple-Input Multiple-Output (MIMO) capabilities. Although grounded in physically meaningful Doppler features, the final decision stage relies on a black-box DL model. We exploit the dataset associated to [14] to generalize the methodology we propose and to compare the results with those obtained on our dataset.

To the best of our knowledge, no prior work has explored activity recognition through an SFT representation in the delay-Doppler domain.

B. SVM-Based CSI Classification

SVMs are extensively used for CSI-based sensing due to their robustness in high-dimensional feature spaces and effectiveness with limited training data. Wang *et al.* [15] demonstrated device-free presence detection and localization using CSI fingerprinting and SVM classifiers, highlighting the discriminating power of CSI amplitude and phase features extracted from Wi-Fi preambles. Zhou *et al.* [16] further applied SVM regression for device-free passive localization: by learning a nonlinear mapping between CSI features and spatial locations, their approach achieved improved localization accuracy compared to linear models, confirming the suitability of kernel-based methods for CSI-driven sensing.

Beyond localization, SVMs have been employed for radio environment awareness. Liu *et al.* [17] investigated indoor interference classification using frequency-domain CSI, employing SVMs to differentiate between heterogeneous interference sources. Similarly, Li *et al.* [18] addressed Line of Sight (LoS)/Non-Line of Sight (NLoS) identification via space-frequency domain CSI features, demonstrating that propagation conditions can be reliably inferred directly from CSI measurements. More recently, Chen *et al.* [19] introduced

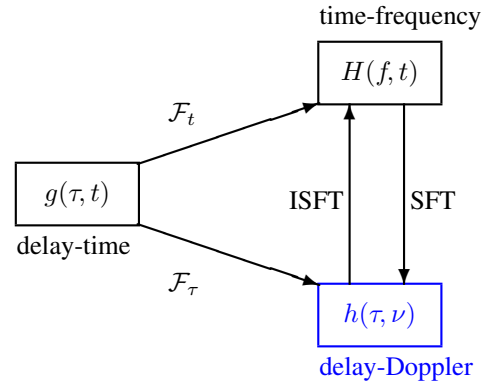


Figure 1: Different domain representations of a time-variant multipath channel impulse response $g(\tau, t)$.

a hybrid SVM-Naive Bayes framework that uses SVMs for NLoS identification prior to device-free positioning. Their results showed that explicitly classifying channel conditions using CSI can significantly improve localization robustness in complex indoor scenarios.

C. Contrast With Deep Learning-Based Methods

In parallel, DL has gained popularity for CSI-based sensing and classification. Deep Neural Networks (DNNs) have been applied to human activity recognition [20], [21], indoor localization [22], [23], and gesture recognition [24]. These methods typically treat CSI time-series representations as images or sequences and rely on large labeled datasets for automated feature extraction; unfortunately, as highlighted in [3], they are often difficult to generalize and reproduce, suggesting that the classification is based on features that are not intrinsic to activity or situation itself.

While DL approaches often achieve superior performance in highly dynamic scenarios, they entail increased computational complexity, reduced interpretability, and substantial training data requirements. In contrast, SVM-based methods offer a lightweight and interpretable alternative, particularly well-suited for scenarios with limited training samples or constrained hardware. Moreover, they allow explicit feature engineering, maintaining closer alignment with the physical interpretation of CSI.

Overall, SVM- and DL-based activity recognition methods represent complementary design points: DL emphasizes representation learning and scalability, whereas SVMs prioritize robustness, efficiency, and physical interpretability.

III. THE SYMPLECTIC TRANSFORM

Multi-domain transformations can be used to find the most suitable representation of a signal to achieve a certain goal. Figure 1 illustrates the multi-domain relationships obtainable through the Fourier Transforms (FTs), with the SFT directly linking the time-frequency and the delay-Doppler domains. A complete theory of the SFT can be found in [25].

In wireless communications, a signal $x(t)$ is transmitted in the time domain and during its propagation it undergoes a distortion that shapes it into the received signal $y(t)$. This distortion is reflected in the CSI, which is computed at the receiver in the frequency domain and used for equalization and sensing. An OFDM-based system that sends N OFDM symbols of duration T (over a session of $N\Delta t$ [s]) transmits

a signal that can be used as a pilot for CSI extraction at the receiver:

$$x_p(t) = \sum_{k=1}^N x(t - k\Delta t) \cdot w_T(t - k\Delta t) \quad (1)$$

where $w_T(t)$ is a unit square window pulse of duration $T < \Delta t$. In other words, the transmitted signal $x_p(t)$ consists of N pilot OFDM symbols (preambles in Wi-Fi) of duration T repeated every Δt . Eq. (1) expresses the nature of the CSI signal in Wi-Fi where a single pilot OFDM symbol of duration T is followed by $(\Delta t - T)[s]$ with the rest of the frame and the silence period separating the frames. All signals we consider in the following are baseband-equivalent, therefore complex-valued.

Given the time-varying channel impulse response $g(\tau, t)$, and assuming $g(\tau, t) \approx g(\tau, k\Delta t)$ for $k\Delta t < t \leq (k+1)\Delta t$, the received signal useful for sensing is

$$\begin{aligned} y_p(t) &= \int_0^{T_c} g(\tau, t) \cdot x_p(t - \tau) d\tau \\ &= \sum_{k=1}^N \int_{k\Delta t}^{k\Delta t + T} g(\tau, k\Delta t) \cdot x_p(t - \tau) d\tau \end{aligned} \quad (2)$$

where T_c is the observation interval.

The Nyquist–Shannon sampling theorem drives the discrete-time processing: the received signal is sampled at $T_s = 1/B$, where B is the channel bandwidth, hence, the channel sensing is limited to a bandwidth B and a configurable duration of the observation T_c . During T_c the signal $x_p(t)$ is bursty, and hence $y_p(t)$ is a sequence of short bursts as defined in Eq. (2).

Segmentation of T_c into N blocks of duration Δt ($T_c = N\Delta t$) gives the discrete samples of the delay-time channel response as a $M \times N$ matrix:

$$H_{DT}[m, n] = g(mT_s, n\Delta t) \quad \begin{cases} m = 0, \dots, M-1 \\ n = 0, \dots, N-1 \end{cases} \quad (3)$$

In packet-based channels, like Wi-Fi, the CSI extraction frequency is related to the packet rate, since the preambles are in the PHY header of the packets. In contrast, in circuit-based channels such as 4G and 5G, it is governed by the pilot repetition rate in the time-frequency matrix of Resource Elements. In Wi-Fi, the pilot symbols transmitted every Δt , have a duration $T \ll \Delta t$ and they correspond to a single OFDM symbol of duration $T = 1/\Delta f$ ($\simeq 4$ or $8 \mu s$ depending on the Wi-Fi version), where Δf is the subcarrier spacing.

Given $g(\tau, t)$, applying the Discrete Fourier Transform (DFT) to the columns of the matrix in Eq. (3) (corresponding to the delay axis τ) we obtain the samples of the frequency-time channel response $H(f, t)$:

$$H_{FT}[m, n] = H(m\Delta f, n\Delta t) \quad \begin{cases} m = 0, \dots, M-1 \\ n = 0, \dots, N-1 \end{cases} \quad (4)$$

By applying the DFT to the rows of the matrix in Eq. (3) (corresponding to the time axis t), we obtain the samples of the delay-Doppler channel response $h(\tau, \nu)$:

$$H_{DD}[m, n] = h(mT_s, n/T_c) \quad \begin{cases} m = 0, \dots, M-1 \\ n = 0, \dots, N-1 \end{cases} \quad (5)$$

The direct transformation from $H(f, t)$ to $h(\tau, \nu)$ is called Symplectic Fourier Transform (SFT), which is invertible, being similar to a two-dimensional FT.

We call the delay-Doppler representation H_{DD} (Eq. (5)) of N time samples and M frequency samples a *cast* of the experiment: this is the elementary piece of information we use for activity recognition.

IV. DELAY-DOPPLER-BASED FEATURE EXTRACTION

The approach proposed in this work leverages the physical interpretation of the CSIs provided by the DD domain, but to accomplish this we need to identify the most representative features of the DD representation.

To avoid any ambiguity, we clarify that the term *experiment* identifies a seamless session of measurement with a person continuously performing the same activity. The diversification across rooms in the two considered datasets ensures that the performance of the proposed method is not related to the circumstances where the CSIs were collected, highlighting the consistency of the methodology across various environments.

For every experiment and room, we apply a sliding window of duration T_c to the CSI time series. Given a sequence of N_c CSI samples, this produces $N_c - N + 1$ partially overlapping windows of length N , for which we compute the corresponding casts H_{DD} . For each cast, we only keep the $M' = 11$ central bins on the delay axis, as they retain the largest amount of information on the multipath components introduced by the propagation environment. As a result, we define the cast as the amplitude of a complex-valued matrix $H_{DD} \in \mathbb{C}^{N \times M'}$, where N depends on the desired window duration and the inter-frame interval Δt . For instance, with inter-frame $\Delta t = 100$ ms, a 3 s cast corresponds to $N = 30$ samples and Fig. 2 shows its heatmap representation.

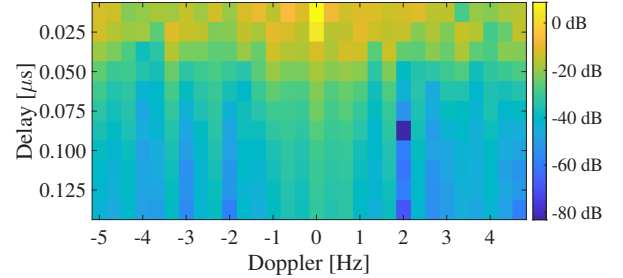


Figure 2: Example of 3 s cast H_{DD} computed on an ANS Walking experiment with inter-frame $\Delta t = 100$ ms.

Since four antennas are enabled on the device extracting the CSI samples, as further clarified in Sect. VI, the DD cast is first computed independently for each antenna and then averaged across the four. This reduces noise and variability while preserving the effects induced by human movements on all receiver chains. The amplitude of the resulting average cast ($\overline{H}_{DD}$) is the basic sensing unit used in the activity recognition task: each cast is labeled with the activity carried out during the experiment it belongs to, and this labeling serves as the ground truth. More sophisticated uses of the multiple antennas will be investigated in future work.

Although the casts are already highly informative and more interpretable than the CSI in the time-frequency domain, they remain high-dimensional objects. To tackle this issue and obtain manageable yet meaningful amounts of information, we extract a compact set of physically interpretable features from each $\overline{H}_{DD}(m, n)$. The features are designed to summarize the most relevant properties of the movements captured by the cast and reflected in the DD-domain energy distribution.

Specifically, we compute the total energy of the cast

$$E_c = \sum_m \sum_n |\overline{H}_{DD}(m, n)|^2,$$

which provides a global measure of the overall signal attenuation of the cast. To characterize how the energy is scattered along the two axes of the DD plane, we consider its profiles along Doppler and delay:

$$E^D(n) = \sum_m |\overline{H}_{DD}(m, n)|^2, \quad E^d(m) = \sum_n |\overline{H}_{DD}(m, n)|^2.$$

From the Doppler energy profile $E^D(n)$, we compute its mean E_μ^D , standard deviation E_σ^D , and skewness E_s^D . These statistics describe how strongly energy levels vary along the Doppler axis: E_μ^D quantifies the variability of the energy values, while E_s^D captures their possible asymmetry. The same set of statistics (E_μ^d , E_σ^d , E_s^d) is computed on the delay energy profile $E^d(m)$, reflecting the effects of both multipath components and the human body on the propagating signal.

Additionally, we compute the Doppler spread σ_n , characterizing how widely the energy is located along the Doppler axis. To this end, the Doppler energy profile is first normalized with respect to its total energy:

$$P_D(n) = \frac{E^D(n)}{\sum_k E^D(k)}$$

to define energy weights over the Doppler coordinate. The Doppler spread is then computed as the second-order central moment of the Doppler coordinate:

$$\bar{n} = \sum_n n P_D(n), \quad \sigma_n = \sqrt{\sum_n (n - \bar{n})^2 P_D(n)}.$$

While E_σ^D measures how the energy values fluctuate in magnitude, σ_n measures how far the Doppler coordinates carrying energy are from their weighted mean.

To encompass the complexity of the patterns in the DD domain, we compute the entropy η of the normalized energy map $P(m, n)$:

$$P(m, n) = \frac{|\overline{H}_{DD}(m, n)|^2}{E_c}$$

$$\eta = - \sum_m \sum_n P(m, n) \log P(m, n);$$

low entropy values correspond to static scenarios, whereas higher entropy is associated to a more complex pattern due to dynamic activities. Finally, we extract the value Pk_v and the delay and Doppler location of the dominant peak, Pk_l , capturing the impact of the component causing the strongest reflection.

The resulting feature vector is defined as:

$$F = \{E_c, E_\mu^D, E_\sigma^D, E_s^D, E_\mu^d, E_\sigma^d, E_s^d, \sigma_n, \eta, Pk_v, Pk_l\},$$

and it provides a compact and interpretable representation of each cast, reducing the cast dimensionality without discarding useful information.

V. AN SVM FOR ACTIVITY RECOGNITION

To perform the activity recognition task, we use an SVM working as a supervised classifier. It is modelled to take as input the feature vectors extracted from the casts collected in the considered room, and learn a decision function that allows to discern between vectors corresponding to different activities represented in the feature space. Thanks to the low dimensionality of the extracted feature vectors, we can adopt a *linear* SVM, which aims to identify the hyperplanes that maximize the margin between vectors labeled as different activities. This approach also minimizes the risk of overfitting associated to more complex nonlinear models.

Since the activity recognition task is not a binary classification problem, we remain compliant with the binary nature of SVMs by implementing a multi-class classification approach through Matlab's `fitcecoc` function [26].

After defining the multi-class SVM model, two evaluation protocols are implemented, as further described in Sect. VIII. In one case, we implement a *k-fold Cross-Validation (CV) strategy*: we train k independent instances of the same SVM classifier on $k - 1$ different subsets (folds) of the input data and test each instance on its remaining unseen fold. The resulting performance metrics are averaged across all k test folds to obtain an estimate of the classifier performance on unseen data. In the other case we perform a *prediction evaluation* through a chronological train-test split of the casts of each experiment: the classifier is trained on the first half of the sequence and evaluated on the second, unseen portion.

VI. DATASETS AND EXPERIMENTAL SETUP

We test the proposed approach on two independent datasets to validate it on experiments performed by different teams with distinct hardware setups and environmental conditions. Both datasets contain CSI's collected throughout several months, with and without LoS between the transmitter and the extractor, and the people performing the activities vary across the experiments. The realistic nature of both datasets helps to show how the approach yields consistent results regardless of the variability of the analyzed data, indicating good adaptability of the methodology. Moreover, since all CSI's were collected without artificially engineering the experimental setup, the datasets reflect real life situations and the extracted information truthfully represents the effects that human movements have on the CSI.

A. ANS Dataset

This dataset consists of experiments performed throughout several months by the Advanced Networking Systems (ANS) research group¹ at the University of Brescia.

The task of collecting CSI samples requires to first identify some boundaries to the otherwise infinite space of experiments. To this goal, we use an everyday office as the reference propagation environment and place the transmitter (Tx) and receiver (Rx) in fixed positions, as displayed in Fig. 3. The Rx is placed on a desk in one corner of the room while the Tx is positioned on a chair behind a drywall partition partially obstructing the LoS (Partial Line of Sight (PLoS)). In the experiments where full LoS obstruction is desired, an aluminum shield is placed on a chair in front of the Tx in position S in Fig. 3.

¹<https://ans.unibs.it>

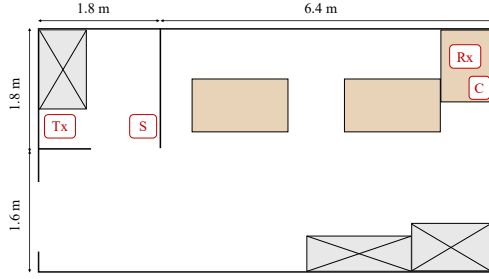


Figure 3: Office floor plan with main furniture, Tx, Rx, and shield positions.

Table I: Number of CSI's in the ANS dataset for each inter-frame value and activity.

Inter-frame	5 ms	10 ms	25 ms	50 ms	100 ms (Beacon)	Total
Empty PLoS	23846	17987	7188	3548	3578	56147
Sitting PLoS	59607	29763	14372	7182	3582	114506
Walking PLoS	33592	17993	7088	3547	3581	65801
Empty NLoS	35756	17981	7088	3546	3652	68023
Sitting NLoS	70454	35706	14295	7191	3554	131200
Walking NLoS	35719	11803	7156	3593	3582	61853
Total	258974	131233	57187	28607	21529	497530

It is important to note that the experimental setup and activities that we consider mimic real-life scenarios, with generated traffic that is fully compliant with the 802.11 standard. The CSI collection is carried out by the Rx using an extension of the Nexmon CSI Extractor tool [27], [28] that allows full customization of the experimental configuration of 802.11ax frames. The Tx and Rx are, respectively, an RT-AX82U IEEE 802.11ax and an RT-AX86U IEEE 802.11ax Access Point (AP) from Asus, both equipped with the Wi-Fi Broadcom chipset BCM43684. The Rx has 4 antennas enabled for CSI extraction, while the Tx only has one transmitting antenna.

This study encompasses three situations labelled as different activities: empty room (E), person sitting (S), and person walking around (W). CSI samples are collected at different frame rates across the experiments ($\Delta t = 5, 10, 25, 50$ ms), to assess the impact of traffic rate on the reachable resolution for activity recognition. CSI samples extracted from Wi-Fi beacons ($\Delta t = 100$ ms) are also considered. The communication takes place on a single spatial stream set up on channel 155 anchored on channel 157 with central frequency of $f_c = 5.775$ GHz and bandwidth of 80 MHz.

The ANS dataset is available at <https://zenodo.org/records/18924941?preview=1> and its structure and content are summarized in Tab. I.

B. SHARP-ax Dataset

To assess the performance of the proposed method across variable sources of CSI samples, we also use the dataset shared by Meneghello *et al.* [29], whose purpose is to provide data to develop activity recognition applications using CSI's extracted from commercial off-the-shelf Wi-Fi devices.

The dataset consists of Wi-Fi CSI's collected through the Nexmon CSI Extraction tool within a single environment. It encompasses three activities – standing in place, walking, and running – together with data from the empty environment.

To align the analysis with that performed on the ANS dataset, we retain only the CSI samples collected in the

empty room or in the presence of a person standing or walking. Standing (S) is adopted as the static human-presence condition, as data collected with a person sitting are not available. Samples extracted during the running activity are excluded to maintain a balanced number of CSI's per activity level and avoid bias towards highly dynamic scenarios.

To collect the dataset, the authors replicated real-life environmental conditions and activities, setting up an IEEE 802.11ax network with two Asus RT-AX86U APs placed 4 m apart along the two longer walls of a house corridor. The devices exchanged Wi-Fi frames inter-spaced by $\Delta t = 7.5$ ms; the communication took place over one spatial stream on channel 157 with 80 MHz bandwidth. The Tx had one enabled antenna, while the Rx had four. Overall, the dataset encompasses 185 747 CSI samples: 63 496 for the empty room, 61 125 for the person standing, and 61 126 for the walking activity.

The LoS conditions where this dataset was collected support the evaluation of the proposed activity recognition method across three obstruction levels of the propagation path: LoS, PLoS, and NLoS.

VII. CSI IN Wi-Fi

While the methodology we propose can be applied to any wireless communication system based on OFDM modulation schemes, in the following we focus on Wi-Fi technology.

Wi-Fi CSI's are extracted from the preambles transmitted in the frame headers: data collection is discrete-time and directly returns $H(m\Delta f, n\Delta t)$ (Eq. (4)) if frames are transmitted at constant frame rate of $\frac{1}{\Delta t}$ frames/s. Specifically, the CSI is extracted from the Short Training Field (STF) or the Long Training Field (LTF) symbols, as shown in Fig. 4.

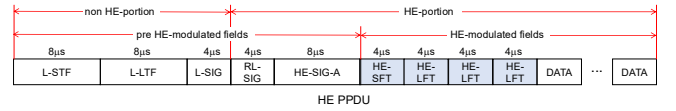


Figure 4: Structure of the Physical Protocol Data Unit (PPDU) frame in Wi-Fi-6 (802.11ax).

Figure 4 describes the structure of a PPDU Wi-Fi-6 High Efficiency (HE) frame, with the fields that are used to compute the CSI highlighted in blue. Legacy symbols are only used for the initial tuning of the CSI extraction but do not contribute to its final value.

The bandwidth supported by Wi-Fi-6 (802.11ax), the standard version used in this work, can range from 20 MHz to 40, 80, up to 160 MHz. The subcarrier spacing Δf is 78.125 kHz, while Δt depends on the frame rate set at the transmitter and is constrained by the maximum supported CSI extraction rate at the receiver.

A. Resolution Bounds

The limitations on the axes resolution apply to any sensing method. Specifically, since $B = M\Delta f$ and $\Delta f = 1/T$, the delay resolution is $\Delta\tau = T_s = 1/B = T/M$ and the delay axis spans the range $[0, T]$ [s]. Since $T_c = N\Delta t$, the Doppler shift resolution is $\Delta\nu = 1/T_c = 1/(N\Delta t)$ and the Doppler axis spans the range $[-\frac{1}{2\Delta t}, \frac{1}{2\Delta t}]$ [Hz]. On the Doppler axis, we can distinguish the locations of scatterers in the room that occur over Δt [ms] with Doppler shifts due to movement with a resolution of $\Delta\nu = 1/T_c$ [Hz].

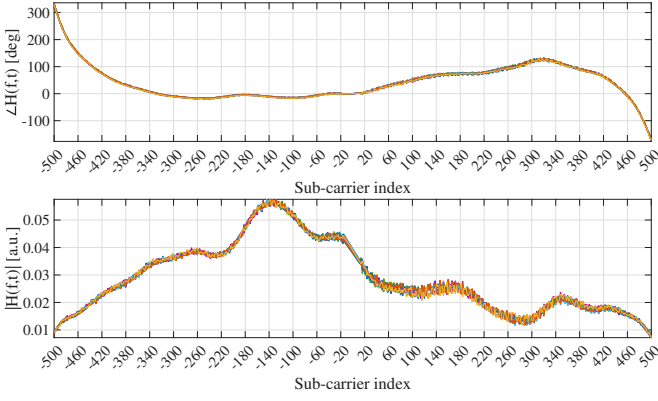


Figure 5: CSI phase and amplitude after data preparation for a group of 10 consecutive frames from the ANS dataset.

Using a bandwidth of 80 MHz, the delay resolution is $\Delta\tau = T_s = \frac{1}{80 \cdot 10^6} = 12.5 \text{ ns}$, corresponding to a spatial resolution $d_s \simeq 3 \cdot 10^8 \cdot 12.5 \cdot 10^{-9} = 3.75 \text{ m}$: given these values, it is very difficult to resolve scatterers and reflectors in indoor environments, even with such large bandwidth.

On the other hand, the Doppler resolution can be increased by choosing an appropriate T_c , keeping in mind that a longer T_c improves Doppler resolution but requires longer observation times. If we take $T_c = 3 \text{ s}$, the Doppler resolution is $\Delta\nu = 0.333 \text{ Hz}$, which, for a central frequency of $f_c = 5.775 \text{ GHz}$, results in a velocity resolution of $\Delta v \simeq 0.0087 \text{ m/s}$.

B. Data Preparation

The CSI extracted by Nexmon is stored as an array of floating point values consisting of the real and imaginary parts of complex numbers; one complex value is extracted for each subcarrier of the channel.

The amplitude values taken by each CSI across all subcarriers are affected by changes in the Automatic Gain Control (AGC) that are not known to the extractor. To compensate for the effect of the AGC, the first data preparation step requires normalizing the energy of each CSI sample.

Next, we need to handle the subcarriers suppressed by the transmitter: Nexmon only returns the CSI for the data subcarriers, while no information is provided for the control subcarriers. Of these, the external null intervals are guard bands and can be discarded, resulting in a slightly reduced channel bandwidth. For 802.11ax, the guard bands involve subcarriers $[-512, \dots, -501]$ and $[+502, \dots, +512]$. The central null interval involving subcarriers $[-2, \dots, +2]$, instead, needs to be reconstructed by interpolation to enable the computation of the SFT, which requires regularly-spaced samples. This reduces the number of available points in the delay domain to $M' = 1001$. The amplitude and phase values on the central subcarriers are linearly interpolated using the values on the two most central data subcarriers.

Third, we need to compensate for the linear phase shift on the subcarriers due to a fixed time delay τ_0 , which can be ignored. To estimate such delay, we fit a linear model $af + b$ over frequency f to the measured unwrapped CSI phases and estimate the slope $a = -2\pi f\tau_0$, yielding $\tau_0 = -\frac{a}{2\pi}$.

Finally, all unwrapped and de-trended CSI phases are aligned at zero on the central subcarrier by forcing $b = 0$ in the linear fit. The effect of the phase and amplitude corrections can be observed in Fig. 5.

VIII. NUMERICAL RESULTS AND DISCUSSION

Due to the casts partially overlapping by construction, casts computed on CSI samples collected within few milliseconds of each other are not statistically independent. Therefore, validation strategies that do or do not account for inter-cast correlation yield different SVM generalization performances. To account for the effects of temporal correlation, we consider two evaluation strategies. The first tests the classifier ability to leverage the features extracted from casts to perform activity recognition; through the second we wish to understand the influence of inter-cast correlation on the classifier performance.

In both strategies, we explore the impact of different cast sizes through sliding windows of size $T_c = 1, 2, 3 \text{ s}$.

A. k -fold: Random Per-Cast Cross-Validation (CV)

The first evaluation strategy shows that the activities can be told apart by interpreting the information provided by the DD representation of the CSI's, validating the meaningfulness of the selected physically explainable features. In other words, the goal is to show that the single cast contains enough information to discriminate the activities, provided that the system is already calibrated on the context of the experiment.

We evaluate the classifier performance through a random k -fold CV strategy ($k = 5$) that creates the training and testing sets by randomly assigning each cast to either. This implies that two consecutive casts may be assigned to the training and testing set respectively; thus, the testing dataset is not entirely independent of the training dataset, since the contents of the two partitions are highly correlated. As a consequence, this strategy introduces information leakage and only provides an optimistic estimate of the average classification performance. For this reason, the accuracy of the k -fold strategy can be considered an upper bound on the average classification accuracy, under the assumption that the SVM was trained on data collected in the same environment as the testing set. It must be noted that this does not imply that the recognition rate for each *individual* activity cannot improve: if the distribution of the input features in the training and testing sets changes, the classifier performance associated to the single activity may surpass this bound.

Indeed, we observe this behavior in Fig. 6, where the classification accuracy for the Empty room and Sitting activity is often higher than the overall average accuracy reported in Tab. II, regardless of the inter-frame value. The lower accuracy associated with the Walking activity is justified by the natural dynamics of a person walking freely in a room: due to the limited area they can move across, there are portions of the room that require to slow down or stop to turn around. During these intervals, the CSI resembles that of a static environment. Additionally, if the person is in an area of the room that is marginal from the perspective of signal propagation (e.g., near the office door), their movements may have limited influence on the multipath structure, further increasing confusion with static activities.

Figure 6 also highlights how the SVM maintains consistent performance across different frame rates, even when using only beacon frames, as shown in the rightmost column of the figure. Since beacons are transmitted over the 802.11ac 20 MHz anchor channel, they only have 53 useful subcarriers for the SFT. Despite the reduced frequency resolution, the SVM classifies each activity with accuracy above 98% in both PLoS and NLoS conditions with 3 s casts. This outcome

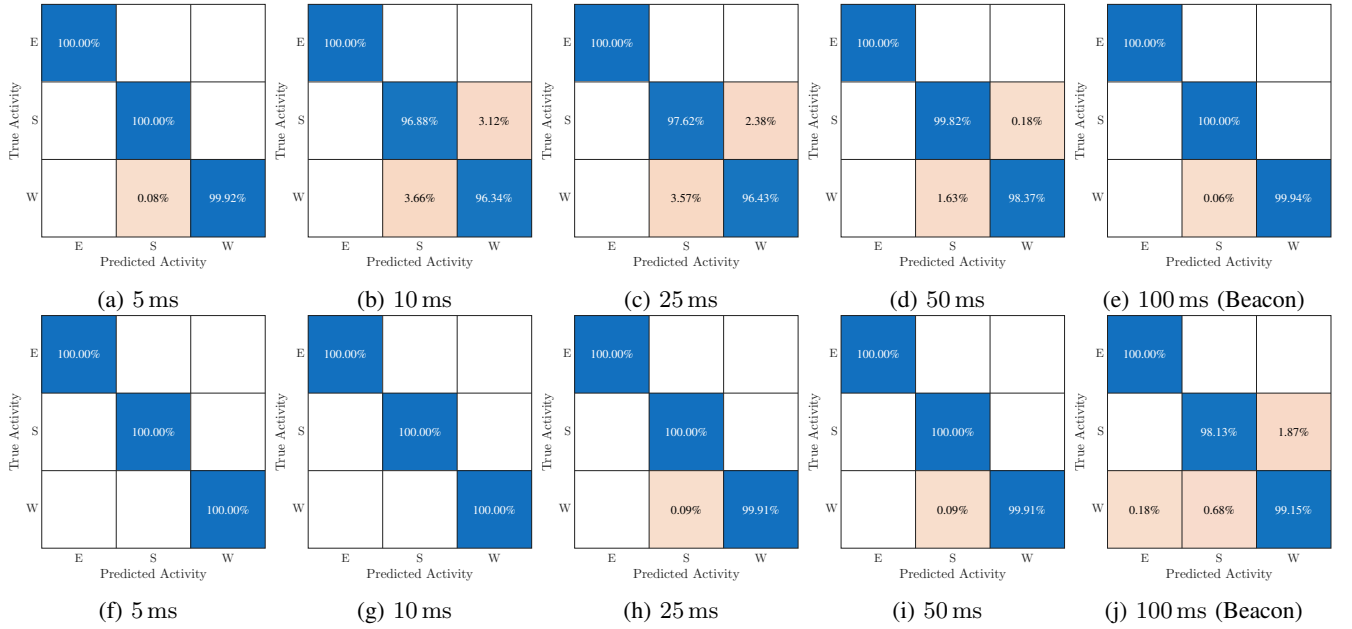


Figure 6: k -fold on ANS dataset: row-normalized confusion matrices showing the percentage of 3 s casts correctly classified for each activity across all inter-frame values. PLoS conditions in the top row, NLoS in the bottom row.

Table II: k -fold on ANS dataset: Average accuracy for the activity recognition task for all inter-frame values as a function of cast duration and LoS obstruction.

Cast Duration	LoS Conditions	5 ms	10 ms	25 ms	50 ms	100 ms (Beacon)
1 s	P	98.41%	95.25%	94.63%	96.17%	98.57%
	N	99.38%	99.72%	98.47%	97.56%	96.99%
2 s	P	99.20%	96.70%	96.90%	98.56%	99.56%
	N	99.75%	100%	99.83%	99.34%	98.50%
3 s	P	99.98%	97.59%	97.92%	99.51%	99.98%
	N	100%	100%	99.98%	99.98%	99.09%

strongly supports the feasibility of true *Joint* Communication and Sensing, achieving high sensing performance by leveraging CSI's extracted from standard communication frames, without needing *ad-hoc* sensing transmissions.

Table II summarizes the average SVM accuracy with the k -fold evaluation on the ANS dataset. Short 1 s casts may not fully capture the effect of a complete movement on the CSI and therefore yield slightly lower accuracies. Conversely, 2 s casts offer a trade-off between observation duration and activity recognition accuracy, showing that beacon-based activity recognition surpasses 98% accuracy without requiring to increase packet rates, which may hinder the system's communication performance.

The same considerations apply to the SHARP-ax dataset (Fig. 7), although the average accuracies are slightly lower than those in Tab. II. This can be explained by comparing the properties of the CSI traces across the two datasets: Fig. 8 shows that the amplitudes of CSI samples in the ANS dataset have very distinct profiles across activities, with variability increasing proportionally to the level of environmental dynamics. Specifically, the Sitting and Walking activities are clearly distinguishable, which simplifies classification for the SVM and causes it to only misclassify few casts from these activities, as already observed in Fig. 6. In contrast, the CSI amplitudes in the SHARP-ax dataset associated with the Standing and Walking activities are much more similar to

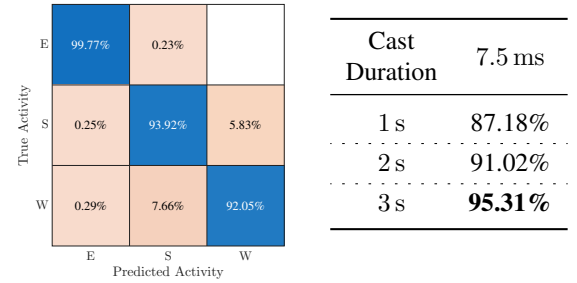


Figure 7: k -fold on SHARP-ax dataset. Left: row-normalized confusion matrix showing the percentage of 3 s casts correctly classified for each activity. Right: average accuracy for the activity recognition task as a function of cast duration.

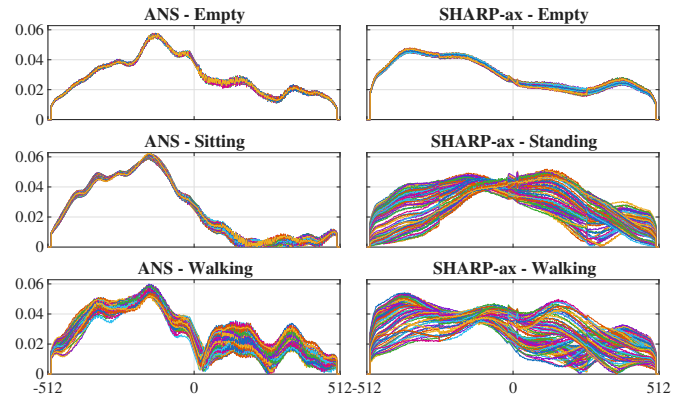


Figure 8: Comparison of 100 CSI samples from experiments in the ANS (left) and SHARP-ax (right) datasets.

each other and therefore easier to misclassify. This causes more frequent confusion between the two labels in the SVM, leading to slightly degraded accuracies, as shown in the confusion matrix on the left in Fig. 7.

However, a comparison of the bold values in Tab. II and Fig. 7 with the results presented in Fig. 2 of [14] clarifies the performance gains achieved by the proposed approach. In particular, the SVM consistently outperforms the SHARP-ax

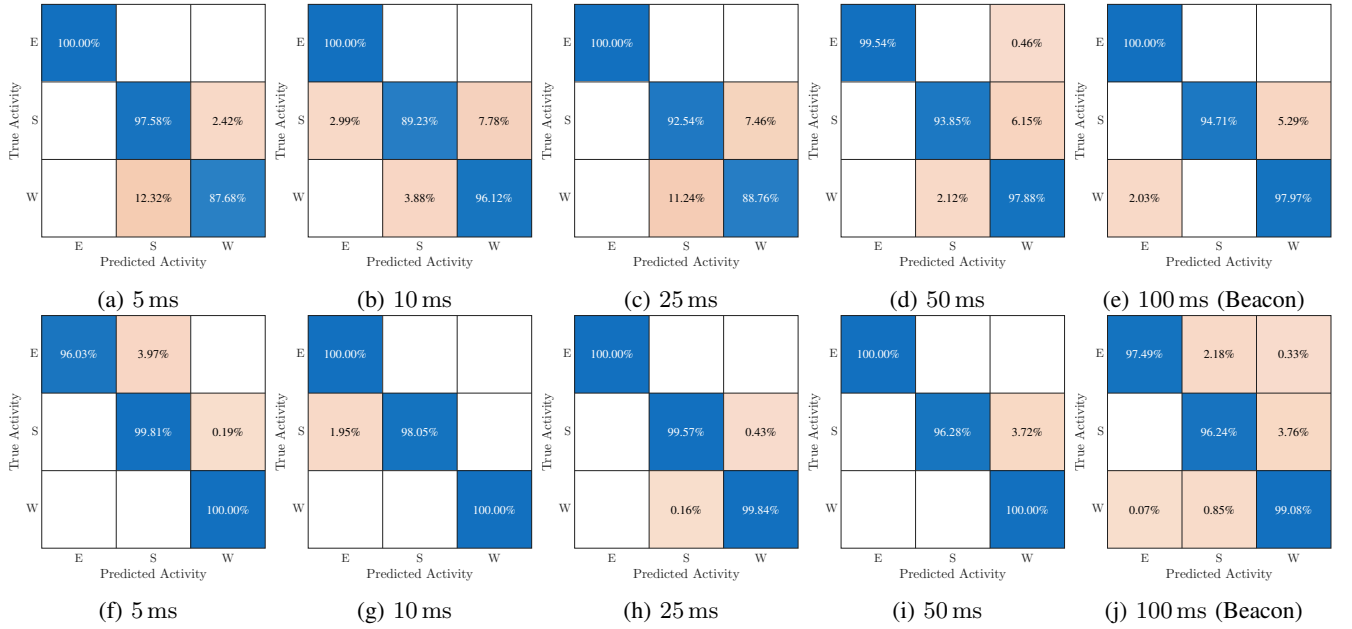


Figure 9: Prediction evaluation on ANS dataset: row-normalized confusion matrices showing the percentage of 3s casts correctly classified for each activity across all inter-frame values. PLoS conditions in the top row, NLoS in the bottom row.

implementation. For 3s casts, the average accuracy in Fig. 7 is noticeably higher than the corresponding average accuracy obtained with Orthogonal Frequency Division Multiple Access (OFDMA) RU1-996 in Fig. 2 of [14]. Data collected over a 20 MHz channel show a similar trend: the average accuracy achieved on the ANS dataset using CSI extracted from Wi-Fi beacons with 3s casts surpasses the average accuracy computed across all RU*-242 in [14]. These results indicate that the proposed method improves classification performance while offering an interpretable alternative to NN-based activity recognition.

B. Prediction Evaluation: Temporally Disjoint Casts

The second evaluation strategy tightens the constraints on how to partition the features dataset to account for the temporal correlation, strengthening the classifier ability to generalize to unseen data. While the activity recognition goal remains the same, it enhances temporal robustness, reducing the optimistic bias of the k -fold CV. This allows us to show that the classifier correctly identifies the activities on future, lower-correlating casts within the same experimental setup, better approximating real deployment conditions.

In this evaluation, every experiment is split so that the feature vectors extracted from the first half of its casts go into the training and the other half into the testing set. This way, training and testing are temporally disjoint and correlation is reduced compared to the k -fold strategy.

As expected, this evaluation is stricter, regardless of the level of LoS obstruction. Table III summarizes the average accuracy reached by the SVM on the ANS dataset as a function of the experimental setup and inter-frame time; the results on the SHARP-ax dataset are shown in Fig. 10. Once again, the SVM maintains consistent performance across different inter-frame values, as shown in Fig. 9, with few misclassifications of the Sitting and Walking activities, due to the reasons already discussed in Sect. VIII-A.

The observed performance reduction is consistent with the expected behavior of a temporally disjoint evaluation. With k -fold, high correlation between training and testing sets boosts

Table III: Prediction evaluation on ANS dataset: Average accuracy for the activity recognition task for all inter-frame values as a function of cast duration and LoS obstruction.

Cast Duration	LoS Conditions	5 ms	10 ms	25 ms	50 ms	100 ms (Beacon)
1 s	P	98.40%	95.16%	88.46%	94.83%	97.00%
	N	97.95%	98.67%	97.91%	96.61%	95.04%
2 s	P	97.56%	95.02%	91.46%	96.39%	98.04%
	N	98.81%	98.83%	98.83%	99.10%	97.37%
3 s	P	96.27%	94.07%	93.48%	96.26%	97.56%
	N	98.90%	98.94%	99.74%	98.13%	97.61%

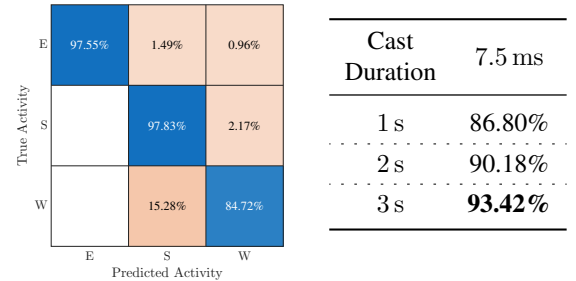


Figure 10: Prediction evaluation on SHARP-ax dataset. Left: row-normalized confusion matrix showing the percentage of 3s casts correctly classified for each activity. Right: average accuracy for the activity recognition task as a function of cast duration.

the recognition rate; by reducing the inter-cast correlation through the 50/50 temporal split, we mitigate this effect, providing a more realistic estimate of the SVM generalization ability. The slightly lower accuracy reflects the removal of the optimistic bias and must not be mistaken for a decreased representativity of the features.

A direct comparison between Tab. II and Tab. III shows that for the ANS dataset the transition from a k -fold to a prediction evaluation consistently reduces the overall accuracy. Averaging across all LoS conditions and inter-frame values, the accuracy loss amounts to 1.515 percentage points for 1s casts, 1.695 percentage points for 2s casts, and 2.305 percent-

age points for 3 s casts. This confirms that the performance of k -fold CV is boosted by the correlation between training and testing sets. When the information leakage is reduced, the SVM solely relies on the intrinsic properties of the feature vectors to recognize the activities, leading to slightly lower but more accurate estimates of its generalization abilities.

The SHARP-ax dataset undergoes the same phenomenon, although the gap between the two evaluation strategies is smaller: the reduction in accuracy is 0.38 percentage points for 1 s casts, 0.84 percentage points for 2 s casts, and 1.89 percentage points for 3 s casts. Unlike the ANS dataset, SHARP-ax maintains accuracy degradation below 2 percentage points for all cast durations. This indicates that the temporal correlation has less impact in this dataset: as illustrated in Fig. 8, consecutive CSI samples in SHARP-ax are less similar to each other than in the ANS dataset both in static and dynamic environments. As a consequence, consecutive casts and their associated feature vectors are less correlated. Hence, the optimistic bias introduced by the inter-cast correlation is smaller, resulting in a less impactful performance degradation when switching to prediction evaluation. Nevertheless, the comparison of the bold values in Tab. III and Fig. 10 with the results presented in Fig. 2 of [14] still highlights that the proposed approach outperforms black-box NN-based methodologies even under stricter constraints on the separation between training and testing sets.

IX. CONCLUSIONS AND FUTURE WORK

This work contributes to the development of explainable methodologies to perform activity recognition. Through the DD-domain representation of the CSI's, we were able to identify a compact set of 11 features that best describe the impact of human motion on the signal's propagation. By feeding these feature vectors to a linear SVM, we implemented a tool capable of discerning with impressive accuracy different levels of dynamics in the propagation environment. The proposed method has low computational complexity, robustness across frame rates and propagation conditions, and full physical interpretability. While also outperforming state-of-the-art NN-based activity recognition methods, the combination of DD-domain feature extraction and a linear SVM classifier poses as a competitive alternative to Artificial Intelligence (AI)-based activity recognition and supports the feasibility of true JCAS.

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